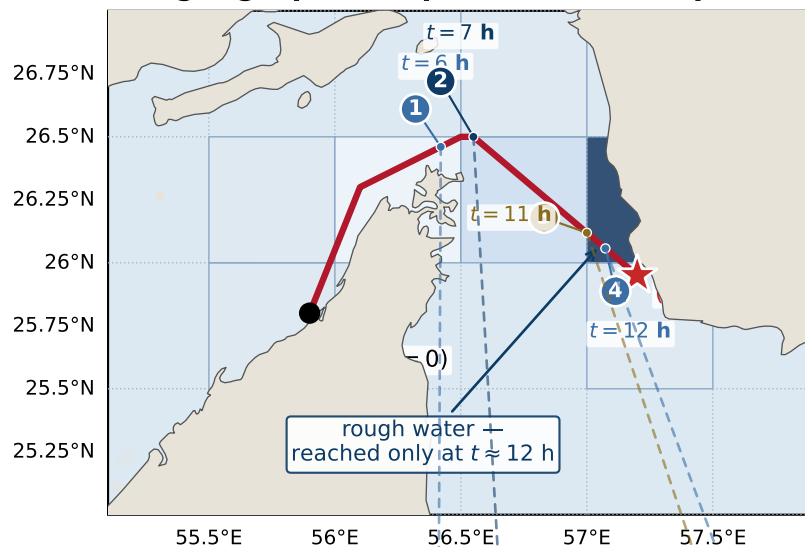


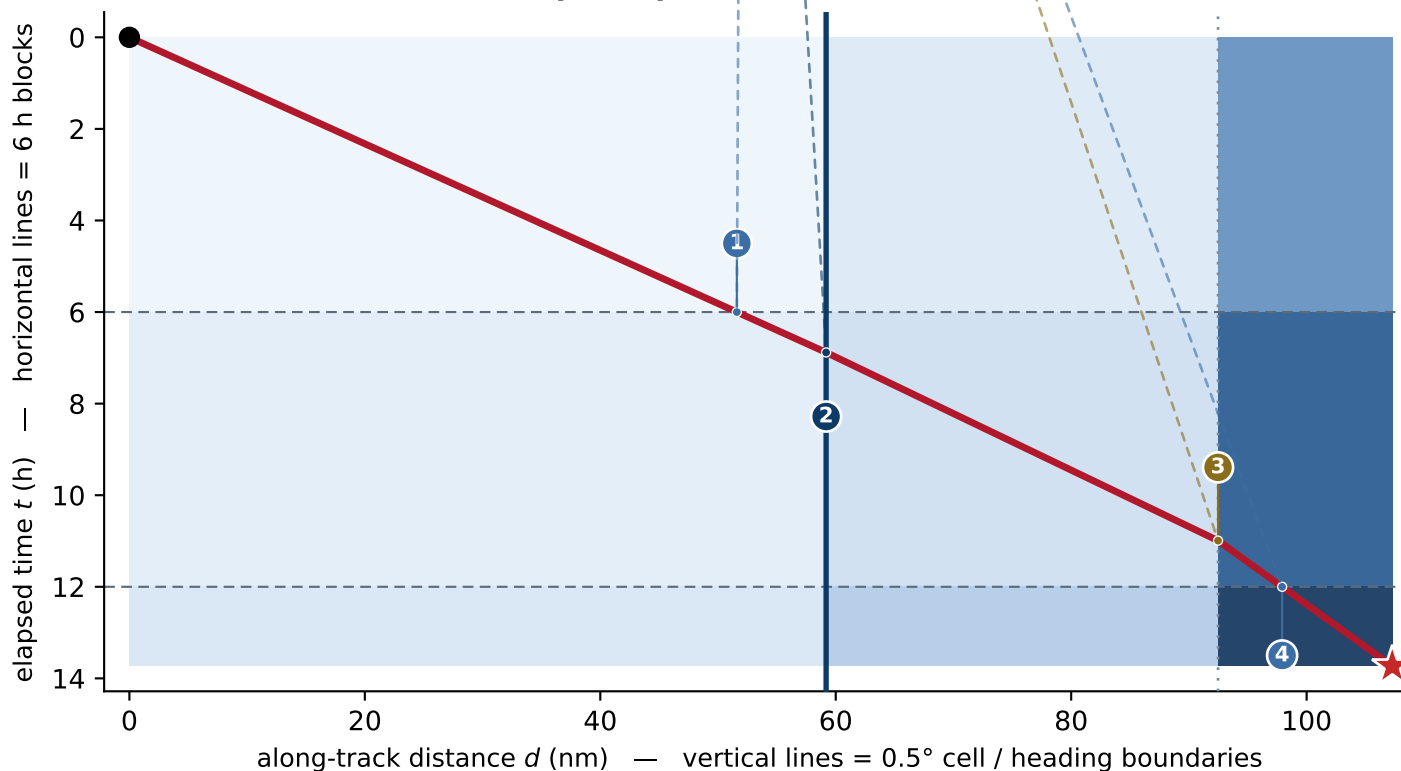
(a) geographic map — where each point is



what each point is

- **Port A**
departure ($t = 0$)
- ① **6 h block**
weather may change (time)
- ② **heading change**
course ψ (segment end)
- ③ **cell crossing**
weather changes (0.5° , space)
- ④ **6 h block**
weather may change (time)
- ★ **Port B**
arrival at the ETA

(b) time $\times$ distance state space, painted with the sea-state field



dashed tie-lines

join each place on the map to WHEN the ship is there — the lower it lands, the later in the voyage

reading the plane (slope = speed)

- flat = fast (calm)
- steep = slow (rough)

